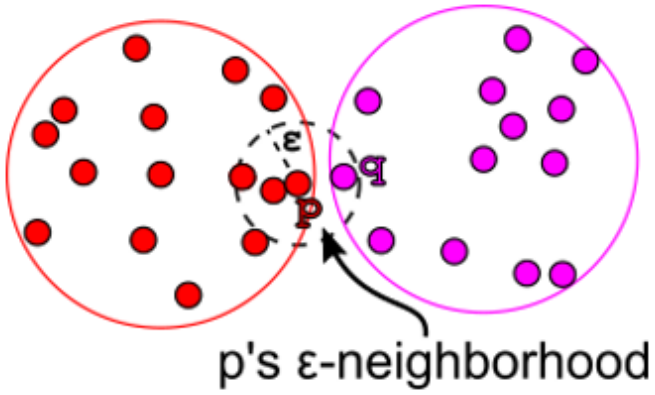


Figure 13.3 Non-overlapping canopy pseudo-clusters could still be considered as belonging to the same cluster by DBSCAN, if both have points close to their border, and the closest points are within each-other ε-distance.

After computing all the clusters for these areas, however, we wouldn't yet be done: since the pseudo-clusters were (possibly) overlapping, the local clusters might also overlap. Besides checking if we should merge clusters that overlap, there is a more subtle effect: as shown in figure 13.3, two non-overlapping clusters could have points that are in each other's ε-neighborhood! So, we need to also check those pseudo-clusters whose hyperspheres are closer to each other than the larger of the values of ε used for those areas (in case DBSCAN was called on them with different values for its hyper-parameters).

The good news is that for such canopies we don't have to check every combination of points drawn from their Cartesian product[5], but only the points in the external ring of each of them, at a distance from the canopy's center greater or equal to $T_2 - \varepsilon$.